

# SULTAN MAHMUD

## Backend Engineer

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## EXPERIENCE

### BACKEND ENGINEER (Freelance/Contract) | Remote

August 2024 - Present

Providing architecture and development expertise for scalable backend systems and cloud-native applications.

- Architected and delivered 5 production systems across **e-commerce** (2 projects), **FinTech** (1), and **SaaS platforms** (2), managing infrastructure serving 5,000+ combined users with 99.9% uptime using **Python** (**FastAPI**, **Django**) and **AWS**.
- Engineered high-concurrency API systems sustaining **1,000+ concurrent** users with **sub-10ms API latency** through Redis caching (**95% hit rate**), **async Python** patterns, and **connection pooling optimization**.
- Implemented **fault-tolerant systems** through strategic implementation of **Redis clustering**, **database replication**, and **automated failover mechanisms**, ensuring **high availability** and **data resilience**.
- Owned full SDLC from architecture design to production deployment: automated infrastructure provisioning of **11+ EC2 instances** across **multi-AZ VPC** using **Pulumi (IaC)** and **Ansible** configuration management, reducing deployment time from 4 hours to 15 minutes with zero-downtime releases via **GitHub Actions CI/CD**.

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### BACKEND DEVELOPER | Cooking Station, Dhaka

June 2024 - August 2024

- Streamlined operations for **200+ users** by designing **role-based admin dashboard** with **real-time meal analytics**, reducing manual subscription **management overhead**
- Eliminated 40% of manual effort** in account management through **automated balance updates** and **error-prevention systems**
- Built production-ready meal scheduling system using **cron jobs** with configurable time boundaries and daily user listing functionality

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## NOTABLE PROJECTS

**STREAMBUDDY: Scalable Video Streaming Platform** *Django REST Framework DASH FFMPEG Celery Redis AWS S3*

**Challenge:** Build adaptive bitrate streaming platform with secure media delivery and automated video processing pipeline

**Impact:** Achieved 60-80% size reduction while maintaining 85-90% quality through multi-resolution encoding with **Nginx**-protected media delivery

#### Technical Achievements:

- Built **Django REST Framework** backend with **dj-rest-auth** authentication, custom throttles (**burst: 60/min**, **streaming: 1000/hour**, **uploads: 10/day**), and comprehensive API documentation via **Swagger/OpenAPI**
- Implemented **asynchronous** video processing pipeline using **Celery** workers with **Redis** queue, **FFMPEG** multi-resolution transcoding (**480p/720p/1080p**) producing **DASH** manifests with **4-second segments** for **adaptive streaming**
- Engineered secure media delivery using **Nginx X-Accel-Redirect** to protect video files from direct access, segment validation for authorized content serving, and **CORS** configuration for cross-origin **DASH** playback
- Integrated **AWS S3** storage with fallback to local filesystem, implemented metadata management for video tracking (status: **UPLOADED**→**QUEUED**→**PROCESSING**→**COMPLETED/FAILED**), and **10GB** upload support with **MIME** validation
- Deployed production infrastructure via **Ansible** automation with **Docker Compose** orchestration, automated **SSH** wait-and-ready checks, and environment template configuration for seamless deployments
- Built comprehensive error handling with custom exceptions (**VideoProcessingError**, **StorageError**, **DuplicateTitleError**), **Celery** task failure handlers updating video status, and **FFmpeg** progress tracking for **real-time** processing feedback

**Key Innovation:** **DASH-based adaptive streaming** with **Nginx**-protected media workflow enabling secure, buffer-free video playback across devices while maintaining **85-90% visual quality** at **60-80% reduced bandwidth**

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**ELASTIKUBE: PRODUCTION K3S AUTOSCALER FOR AWS** | *AWS Lambda EventBridge DynamoDB K3s DynamoDB Prometheus Python Prophet ML Ansible* | [Demo](#)

**Challenge:** Build production-grade intelligent autoscaler with ML-based predictive scaling for K3s clusters on AWS

**Impact:** Event-driven Lambda architecture with 4-layer intelligent scaling achieving 60% cost reduction (\$200→\$111-\$149/month), multi-AZ high availability, and proactive pre-scaling before traffic spikes

#### Technical Achievements:

- Architected **4-layer intelligent autoscaling**: (1) **Data Collection**, (2) **Time-Aware Scaling** (peak/off-peak: 85%/60% vs 60%/40%), (3) **Flash Sale Detection** (>30% spike in 2min), (4) **Prophet ML forecasting CPU 15min ahead**
- **Built ML training pipeline** with **Kubernetes CronJob** for weekly automated retraining, **feature engineering** (cyclical encoding, lag features), **cross-validation**, and **backtesting**
- **Implemented distributed locking** via **DynamoDB conditional writes**, **Write-Ahead Log (WAL)** for crash recovery, and **EventBridge orchestration** for fault-tolerant scaling
- Engineered **multi-AZ high availability** with round-robin distribution across 3 AZs, **LIFO scale-down** with **kubectl drain** via **SSM**, and single **NAT Gateway** optimization
- Deployed comprehensive **observability**: 17 **CloudWatch alarms** (CRITICAL/WARNING), **Prometheus graceful degradation**, fixed **LogGroups**, **DLQ monitoring**, and **SNS alerting**
- Built **spot instance support** with automatic **On-Demand fallback** handling 3 **AWS capacity exceptions** for **70% cost savings**

**Key Innovation:** Hybrid reactive+proactive scaling combining time-aware thresholds, flash sale detection, and ML forecasting for SLA-driven pre-scaling

## TECHNICAL SKILLS

- **LANGUAGE & FRAMEWORKS:** Python, Golang, JavaScript, FastAPI, Django, DjangoREST
- **CLOUD & INFRASTRUCTURE:** AWS(EC2, S3, VPC, Lambda, EventBridge, CloudWatch, SNS, SQS), Docker, Kubernetes(K3s), Pulumi, Ansible
- **DEVOPS & CI/CD:** GitHub Actions, Docker Compose, Nginx, Load Balancing
- **ARCHITECTURE & DESIGN:** Microservices, Distributed Systems, High Availability, Fault Tolerance, Load Balancing, Caching Strategies, Database Optimization
- **DATABASE & CACHING:** PostgreSQL, Redis, MongoDB, SQLAlchemy, PgBouncer (Connection Pooling), Alembic (Migrations)
- **API DEVELOPMENT:** Swagger/OpenAPI, RESTful API, Authentication (JWT), Rate Limiting, CORS
- **MESSAGE QUEUING & ASYNC:** Celery(Worker, Beat, Flower), Redis, RabbitMQ, Async/Await
- **OBSERVABILITY & MONITORING:** OpenTelemetry (Distributed Tracing), Prometheus, Grafana, Tempo

## EDUCATION

### BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING

Daffodil International University, Dhaka

September 2017 - December 2022

## EXTRACURRICULAR ACTIVITIES & COMMUNITY ENGAGEMENT

- **Quora Bengali:** Maintain A Dedicated Blog On **Quora** Offering Insights And Solutions To Inquiries Spanning Technology, Culinary Arts, And Health In Bengali. Accumulated **Readership** Nearing **200,000** With Nearly **200 Followers**.
- **Medium Technical Blog:** Published 2 in-depth technical articles (1,000+ words each) discussing distributed systems, backend architecture & engineering decisions and trade-offs
- **YouTube Channel:** Founded And Managed The Youtube Channel "**I.T. Darshonik**" Producing Over **40 Instructional Videos** In Bengali, Aimed At Bridging The Language Gap In Tech Education.

## AWARDS & ACHIEVEMENTS

- **DIU Take-Off Programming Contest, Spring 2018:** Ranked **6th/300+ participants** (Top 2%)
- **Competitive Programming:** Solved 500+ problems across BeeCrowd, LightOJ, HackerRank, LeetCode.
- **Proficiency:** Dynamic Programming, Graph Theory, Greedy Algorithms, Binary Search